CRSP U.S. stock database - v2021.12

This folder is designed to store the CRSP U.S. Stock flat files for R released by CRSP in December 2021. It also provides code to generate derived data.

Prerequisites

Download the CRSP U.S. Stock flat files for R (`sfz_*.rds` files) released by CRSP in December 2021 and place them into this folder.

Installation

Install the dependencies with renv:

```
renv::restore()
```

Usage

Generate the following data files by running the corresponding R script.

File `frachl.csv.gz`

```
Rscript frachl.R
```

Contains the fractions of days where the open or close prices match the high or low prices.

Field	Type	Description
`KYPERMNO`	INTEGER	Unique permanent security identification number assigned by CRSP to each security
`YYYYMM`	INTEGER	Year-month in the format `YYYYMM`
`FRACOH`	NUMERIC	Fraction of days where the opening price is the highest price
`FRACOL`	NUMERIC	Fraction of days where the opening price is the lowest price
`FRACCH`	NUMERIC	Fraction of days where the closing price is the highest price
`FRACCL`	NUMERIC	Fraction of days where the closing price is the lowest price

File `spread.csv.gz`

```
Rscript spread.R
```

Contains signed bid-ask spread estimates obtained with various methods. A value of 0.01 corresponds to a spread of 1%.

Field	Type	Description
`KYPERMNO`	INTEGER	Unique permanent security identification number assigned by CRSP to each security
`YYYYMM`	INTEGER	Year-month in the format `YYYYMM`
`EDGE`	NUMERIC	Effective spread obtained with the efficient estimator described in this paper
`OHL`, `CHL`, `OHLC`, `CHLO`	NUMERIC	Effective spread obtained with the estimators described in this paper
`AR`	NUMERIC	Effective spread obtained with the estimator described in Abdi & Ranaldo (2017)
`CS`	NUMERIC	Effective spread obtained with the estimator described in Corwin & Shultz (2012)
`ROLL`	NUMERIC	Effective spread obtained with the estimator described in Roll (1984)
`QS`	NUMERIC	Average end-of-day quoted spread within the month